NAME: _____ Geometry (FL B.E.S.T.)

1.2 Measuring and Constructing Segments

Segment length is a number, not just a picture — and the Segment Addition Postulate and the Midpoint definition turn that number into an algebra tool. This section goes past single-step "add the two pieces" problems into equations with a variable on both sides, midpoints of midpoints, and word problems where you must first decide which points are the endpoints before any arithmetic happens. Expect to justify each step (postulate or definition) as much as to compute the final number.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. A at -2, B at 6. Find AB.

2. B between A and C, $AB = 5$, $BC = 9$. Find AC.

3. $AC = 25$, $AB = 11$, B between A and C. Find BC.

4. M midpoint of $\overline{AB}$, $AM = 7$. Find AB.

5. $PM = 2x + 3$, $MQ = 4x - 5$, M midpoint. Find x.

6. $\overline{RS} \cong \overline{TU}$ and $RS = 15$. Find TU.

7. $AB = x$, $BC = 2x + 1$, $AC = 16$, B between. Find x.

8. A at 3, B at 11. Midpoint M?

9. A at 1, midpoint M at 5. Find B.

10. P at -8, Q at -1. Find PQ.

11. $AB = 3x - 1$, $BC = x + 5$, $AC = 20$. Find x.

12. $MQ = 6$, M midpoint of $\overline{PQ}$. Find PQ.

13. $AM = 5x - 2$, $MB = 3x + 6$. Find x.

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14. Trail markers: A at mile 3, D at mile 27. B is the midpoint of $\overline{AD}$, and C is the midpoint of $\overline{BD}$. Find B, C, and BC.

15. B is between A and C. $AB = 2x + 3$, $BC = 3x - 1$, and $AC = 32$. Find x, then find AM where M is the midpoint of $\overline{AC}$.

✓ Answer Key

1. 8
2. 14
3. 14
4. 14
5. $x = 4$
6. 15
7. $x = 5$
8. 7
9. 9
10. 7
11. $x = 4$
12. 12
13. $x = 4$
14. $B = 15, C = 21, BC = 6$
15. $x = 6$; $AC = 32$; $AM = 16$